Operating Systems Lab

(CIC-353)

submitted in partial fulfillment of the
requirement for the award of the degree of

Bachelor of Technology in Information Technology

Submitted by

Nitin Kumar (51)
13515003124

Under the supervision of

DR. Preeti Rathee Professor

DEPARTMENT OF INFORMATION TECHNOLOGY
MAHARAJA SURAJMAL INSTITUTE OF TECHNOLOGY
C4, JANAKPURI, NEW DELHI – 110058
(GURU GOBIND SINGH INDRAPRASTHA UNIVERSITY, DWARKA, NEW DELHI)

Vision of the Institute

To become one of the most admired centres of academic excellence in the field of Engineering & Technology for allround professional development of students to enable them to meet the growing technological needs of the country.

Mission of the Institute

Developing new paradigm in imparting education in the fields of Engineering and Technology and to imbibe national values leading to student empowerment, with a view to prepare them to meet the national and global challenges.

Vision of the Department

To build a culture of innovation and research in students and make them capable to solve upcoming challenges of human life using computing.

Mission of the Department

M1. To develop 'educational pathways' so that students can take their career towards success.

M2. To imbibe curiosity and support innovativeness by providing guidance to use thetechnology effectively.

M3. To inculcate management skills, integrity, honesty and human values through curricular, co-curricular and extracurricular activities.

Program Educational Objectives

PEO1. Graduates of IT program are prepared to be employed by IT industries and be engaged in learning, understanding and applying new ideas.

PEO2. The graduates are prepared to perform effectively as individuals and team members in the workplace, growing into highly technical or project management and leadership roles.

PEO3. Graduates are prepared to apply basic principles of practices of computing grounded in mathematics and science for successfully completing software related projects to satisfy customer business objectives and productively engage in research.

PEO4. Graduates are prepared to pursue higher studies so that they can contribute to the teaching profession, research & development of information technology and other allied fields.

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Program 1 : WAP to implement CPU scheduling for First Come First Serve Algorithm.

Aim : To implement the First Come First Serve (FCFS) CPU Scheduling Algorithm and calculate the Completion Time (CT), Turnaround Time (TAT), Waiting Time (WT), Average Turnaround Time, and Average Waiting Time for the given processes.

Code:-

```
#include <iostream>
#include <vector>
using namespace std;

int main() {
   int n;
   cout << "Enter number of processes: ";
   cin >> n;

   vector<int> pid(n), at(n), bt(n), ct(n), wt(n), tat(n);

   for (int i = 0; i < n; i++) {
      pid[i] = i + 1;

      cout << "\nEnter details for P" << pid[i] << ":\n";
      cout << "Enter Arrival Time: ";
      cin >> at[i];

      cout << "Enter Burst Time: ";
      cin >> bt[i];
   }

   for (int i = 0; i < n - 1; i++) {
      for (int j = i + 1; j < n; j++) {
         if (at[i] > at[j]) {
            swap(at[i], at[j]);
            swap(bt[i], bt[j]);
            swap(pid[i], pid[j]);
```

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```
        }
    }
}

int time = 0;

for (int i = 0; i < n; i++) {
    if (time < at[i])
        time = at[i];

    ct[i] = time + bt[i];
    tat[i] = ct[i] - at[i];
    wt[i] = tat[i] - bt[i];

    time = ct[i];
}

cout << "\nProcess\tAT\tBT\tCT\tTAT\tWT\n";

float avgWt = 0, avgTat = 0;

for (int i = 0; i < n; i++) {
    cout << "P" << pid[i] << "\t"
         << at[i] << "\t"
         << bt[i] << "\t"
         << ct[i] << "\t"
         << tat[i] << "\t"
         << wt[i] << endl;

    avgWt += wt[i];
    avgTat += tat[i];
}

cout << "\nAverage Waiting Time: " << avgWt / n;
cout << "\nAverage Turnaround Time: " << avgTat / n;

return 0;
```

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}

Output :-

```
PS C:\Users\student\Desktop\OS_Exper\nitin_Kumar

Enter number of processes: 5

Enter details for P1:
Enter Arrival Time: 0
Enter Burst Time: 5

Enter details for P2:
Enter Arrival Time: 2
Enter Burst Time: 3

Enter details for P3:
Enter Arrival Time: 4
Enter Burst Time: 1

Enter details for P4:
Enter Arrival Time: 1
Enter Burst Time: 4

Enter details for P5:
Enter Arrival Time: 6
Enter Burst Time: 2

Process AT      BT      CT      TAT     WT
P1      0       5       5       5       0
P4      1       4       9       8       4
P2      2       3       12      10      7
P3      4       1       13      9       8
P5      6       2       15      9       7

Average Waiting Time: 5.2
Average Turnaround Time: 8.2
```

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Program 2: WAP to implement CPU scheduling for Shortest Job First Algorithm.

Aim : To implement the Shortest Job First (SJF) CPU Scheduling Algorithm and calculate the Completion Time (CT), Turnaround Time (TAT), Waiting Time (WT), Average Turnaround Time, and Average Waiting Time for the given processes.

Code:

```
#include <iostream>
#include <vector>
#include <algorithm>
using namespace std;

struct Process {
   int pid, at, bt, ct, tat, wt;
};

int main() {
   int n;
   cout << "Enter no. of process : ";
   cin >> n;

   vector<Process> p(n);
   for (int i = 0; i < n; i++) {
      p[i].pid = i + 1;
      cout << "\nEnter Details for the process " << p[i].pid << " :\n";
      cout << "Enter Arrival Time of process " << p[i].pid << ": ";
      cin >> p[i].at;
      cout << "Enter Burst Time of process " << p[i].pid << ": ";
      cin >> p[i].bt;
   }

   int time = 0, completed = 0;
   vector<bool> done(n, false);
   while (completed < n) {
      int index = -1;
      int shortest = 1000000;
      for (int i = 0; i < n; i++) {
         if (!done[i] && p[i].at <= time && p[i].bt < shortest) {
```

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```
            shortest = p[i].bt;
            index = i;
          }
        }

        if (index == -1) {
          time++;
          continue;
        }
        time += p[index].bt;
        p[index].ct = time;
        p[index].tat = p[index].ct - p[index].at;
        p[index].wt = p[index].tat - p[index].bt;

        done[index] = true;
        completed++;
      }
      double avgWt = 0, avgTat = 0;

      cout << "\nP    AT   BT   CT   TAT   WT\n";

      for (int i = 0; i < n; i++) {
        cout << "P" << p[i].pid << "    "
            << p[i].at << "    "
            << p[i].bt << "    "
            << p[i].ct << "    "
            << p[i].tat << "    "
            << p[i].wt << endl;

        avgWt += p[i].wt;
        avgTat += p[i].tat;
      }

      cout << "\nAverage waiting time : " << avgWt / n;
      cout << "\nAverage turn around time : " << avgTat / n;

      return 0;
}
```

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OUTPUT:

```
PS C:\Users\student\Desktop\OS_Exper\nitin_Kumar

Enter no. of process : 5

Enter Details for the process 1 :
Enter Arrival Time of process 1: 0
Enter Burst Time of process 1: 7

Enter Details for the process 2 :
Enter Arrival Time of process 2: 1
Enter Burst Time of process 2: 3

Enter Details for the process 3 :
Enter Arrival Time of process 3: 2
Enter Burst Time of process 3: 5

Enter Details for the process 4 :
Enter Arrival Time of process 4: 3
Enter Burst Time of process 4: 2

Enter Details for the process 5 :
Enter Arrival Time of process 5: 4
Enter Burst Time of process 5: 4

P    AT   BT   CT   TAT   WT
P1   0    7    7    7     0
P2   1    3    10   9     6
P4   3    2    12   11    9
P5   4    4    16   12    8
P3   2    5    21   19    14
```

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